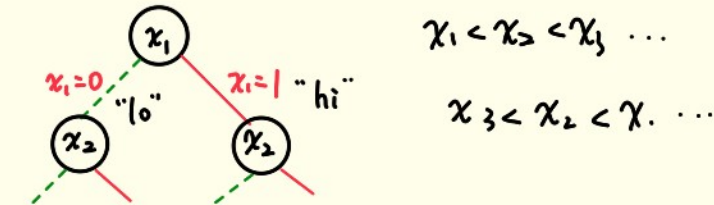


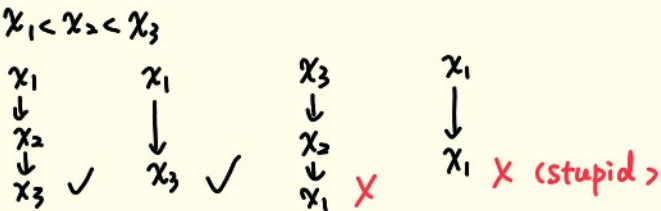
Notes taken by Yucheng Liu, undergraduate at Peking University

19 Sep. 2026 **Lecture 3 Binary Decision Diagram (BDD 二元决策图)** *Powerful!*

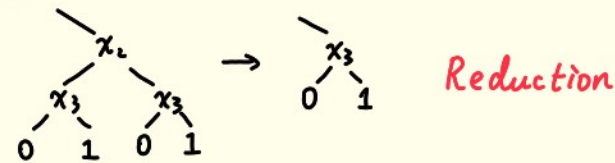
Turn a truth table into a decision diagram
Dotted green line: 0
Solid red line: 1



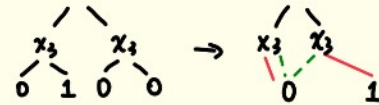
orders can change its look -> this could be a problem -> this decision diagram is not **UNIQUE**
we need a **Canonical Form** -> restrict global ordering of variables



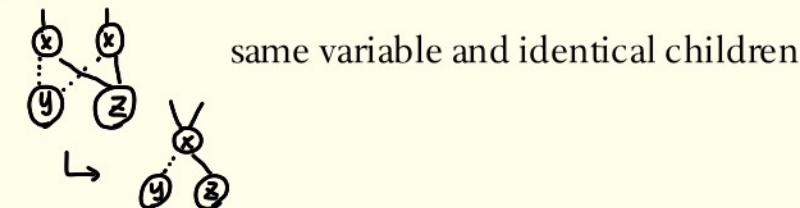
1 Delete meaningless nodes



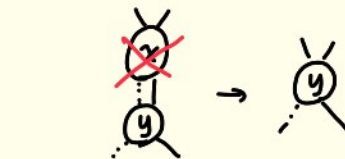
2 merge equivalent leaves



3 merge isomorphic nodes



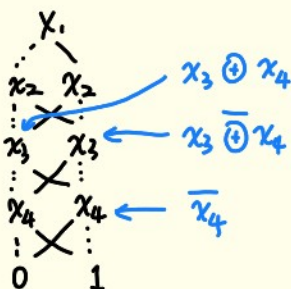
4 eliminate redundant tests



So this is **Reduced Ordered BDD (ROBDD)**
And it's a **Canonical Form** (data structure) for any boolean function!

You can represent any function as a ROBDD

Odd Parity:



Every BDD node (not just root) represents some Boolean function in a **canonical** way

20 Sep. 2026

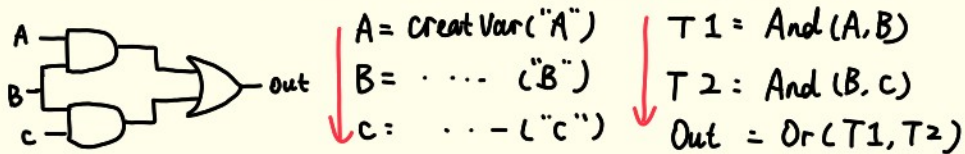
BDD sharing: multi-rooted BDD

How is it built: **Recursive** method, like **URP**

Big trick: Implement each op so if inputs **shared, reduced, ordered** -> outputs are too



AND OR NOT XOR CoFactor $\forall$ Quant $\exists$ Quant Satisfy etc



Applications

is $F == G$?

- Build their BDDs; compare pointers to roots of F, G ; if pointers are same, $F == G$
- Can you find an input making them different?
- Build their BDDs, build the BDD for $H = F \oplus G$; ask SAT for H

Tautology:



Satisfiability:



But ordering matters: Nothing is perfect

Tiny change of the order leads to complexity. And there's no universal solution. "Multipliers are terrible" -> even the best order is exponential

Intuition:

- related inputs should be near each other
- groups of inputs that can determine function by themselves should be closer and near the top of BDD

Prof. Rutenbar is quite cute here lol.

- many chain circuits: Adders Subtractors, Comparators, Priority encoders
a0 b0 a1 b1 a2 b2 ... an bn